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United States Patent
Kind Code
Date of Patent
Inventor(s)

12414842
B2
September 16, 2025
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Creating a digital dental model of a patient's teeth using interproximal information

Abstract

Creating a digital tooth model of a patient's tooth using interproximal information is provided. Interproximal information is received that represents a space between adjacent physical teeth of the patient. A digital teeth model of a set of physical teeth of the patient that includes the adjacent physical teeth is received. One or more digital tooth models is created that more accurately depicts one or more of the physical teeth than the corresponding digital teeth included in the digital teeth model based on the interproximal information.

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Appl. No.: 18/601762
Filed: March 11, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240299133 A1	Sep. 12, 2024

Related U.S. Application Data

continuation parent-doc US 17977852 20221031 US 11957532 child-doc US 18601762
continuation parent-doc US 16846118 20200410 US 11534266 20221227 child-doc US 17977852
continuation parent-doc US 13786300 20130305 US 10617489 20200414 child-doc US 16846118
us-provisional-application US 61739600 20121219

Publication Classification

Int. Cl.: A61C7/00 (20060101); A61C9/00 (20060101)

U.S. Cl.:

CPC A61C7/002 (20130101); A61C9/0053 (20130101);

Field of Classification Search

CPC: A61C (7/002); A61C (9/0053)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Patent Application No. U.S. Ser. No. 17/977,852 filed Oct. 31, 2022, now U.S. Pat. No. 11,957,532, issued on Apr. 16, 2024, which is a continuation of U.S. patent application Ser. No. 16/846,118, now U.S. Pat. No. 11,534,266, issued on Dec. 27, 2022, which is a continuation of U.S. patent application Ser. No. 13/786,300, now U.S. Pat. No. 10,617,489, issued on Apr. 14, 2020, which claims priority to and benefit of U.S. Patent Application No. 61/739,600, filed on Dec. 19, 2012, the disclosures of each of which are incorporated herein by reference in their entirety. (2) The subject matter of this application is related to that of U.S. Pat. No. 10,098,714, issued on Sep. 26, 2018, and any related applications and patents issued therefrom, all assigned to the assignee of the present application and to the extent not repeated herein, the disclosures of each of which are incorporated herein by reference in their entirety. (3) The subject matter of this application is related to that of U.S. Pat. No. 9,668,829, issued on May 17, 2017, and any related applications and patents issued therefrom, all assigned to the assignee of the present application and to the extent not

repeated herein, the disclosures of each of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Orthodontic treatments involve repositioning misaligned teeth and improving bite arrangements for improved cosmetic appearance and dental function. Conventionally, repositioning of teeth has been accomplished by what are commonly referred to as “braces.” Braces comprise a variety of elements such as brackets, bands, archwires, ligatures, and O-rings. After some of these elements are bonded to the teeth, periodic appointments with the treating doctor are required to adjust the braces. This involves bending or installing different archwires having different force-inducing properties, and/or replacing ligatures and O-rings.

(2) An alternative to braces includes the use of elastic positioning dental appliances (also known as “aligners”) for repositioning teeth. Such an appliance can be comprised of a thin shell of elastic material that generally conforms to a patient's teeth but each appliance to be used at a treatment stage has a cavity geometry that is slightly out of alignment with the teeth arrangement at the start of that treatment stage. Placement of the elastic positioning dental appliance over the teeth applies controlled forces in specific locations to gradually move the teeth into a new arrangement as defined by the cavity of the appliance. Repetition of this process moves the teeth through a series of intermediate arrangements to a final desired arrangement. Due to the limited space within the oral cavity and extensive movements that some teeth typically undergo as a part of treatment, the teeth will often be moved throughout the series of intermediate tooth arrangements to properly arrange the teeth. Thus, a single patient treated with elastic positioning dental appliance may experience from 2 to perhaps 50 or more aligner stages (with an average of 25-30 aligner stages per arch) before achieving the final desired teeth arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and form a part of this Description of Embodiments, illustrate various embodiments of the present invention and, together with the description, serve to explain principles discussed below:

(2) FIG. 1 illustrates an example of a patient's physical teeth, according to one embodiment.

(3) FIG. 2 illustrates an example of a digital teeth model that represents the patient's set of physical teeth, according to one embodiment.

(4) FIG. 3 illustrates an example of a conventional three dimensional (3D) virtual model that was created based on a digital teeth model, according to one embodiment.

(5) FIG. 4 illustrates the use of an instrument to measure an interproximal space between adjacent physical teeth, according to one embodiment.

(6) FIG. 5 illustrates an example of a user interface, according to one embodiment.

(7) FIG. 6 illustrates an example of created digital tooth models, according to one embodiment.

(8) FIG. 7 illustrates a system for creating a digital tooth model of a patient's tooth using interproximal information, according to one embodiment.

(9) FIG. 8 illustrates a method of creating a digital tooth model of a patient's tooth using interproximal information, according to one embodiment.

(10) FIGS. 9a-9f depict scannable objects that can be inserted between adjacent physical teeth and used as a part of determining interproximal information that represents a space between the adjacent physical teeth, according to various embodiments.

(11) FIGS. 10a-10c depict scannable objects inserted between adjacent physical teeth, according to various embodiments.

(12) FIG. 11 depicts a digital tooth that represents one tooth that may be used to match or closely

approximate the position of a corresponding tooth, according to one embodiment.

(13) FIGS. 12A and 12B depict an example of a patient's physical teeth at various stages of position as a part of performing interproximal reduction, according to various embodiments.

(14) The drawings referred to in this Brief Description should not be understood as being drawn to scale unless specifically noted.

DESCRIPTION OF EMBODIMENTS

(15) Reference will now be made in detail to various embodiments of the subject matter, examples of which are illustrated in the accompanying drawings. While various embodiments are discussed herein, it will be understood that they are not intended to limit to these embodiments. On the contrary, the presented embodiments are intended to cover alternatives, modifications and equivalents, which may be included within the spirit and scope of the various embodiments as defined by the appended claims. Furthermore, in the following Description of Embodiments, numerous specific details are set forth in order to provide a thorough understanding of embodiments of the present subject matter. However, embodiments may be practiced without these specific details. In other instances, well known methods, procedures, components, and circuits have not been described in detail as not to unnecessarily obscure aspects of the described embodiments.

(16) Unless specifically stated otherwise as apparent from the following discussions, it is appreciated that throughout the description of embodiments, discussions utilizing terms such as “receiving,” “accessing,” “creating,” “depicting,” “specifying,” “obtaining,” “representing,” “corresponding,” “including,” “identifying,” “removing,” “moving,” “determining,” or the like, refer to the actions and processes of a computer system, data storage system, storage system controller, microcontroller, processor, or similar electronic computing device or combination of such electronic computing devices. The computer system or similar electronic computing device manipulates and transforms data represented as physical (electronic) quantities within the computer system's/device's registers and memories into other data similarly represented as physical quantities within the computer system's/device's memories or registers or other such information storage, transmission, or display devices.

(17) Various methods can be used for creating digital teeth models that can be used for record keeping/visualization, restorative or orthodontic purposes. One of the orthodontic purposes which digital teeth models can be used is to help create dental appliances for correcting the position of a patient's teeth. One method involves making a three dimensional (3D) virtual model of the patient's physical teeth. The three-dimensional virtual model can be based on a digital directly or indirectly of a patient's teeth such as directly by an intraoral scan or other direct scan of a patient's physical teeth or indirectly based on a scan of a manually obtained impression of the patient's physical teeth or a scan of a model made from a manually obtained impression of the patient's physical teeth. All of these are collectively referred to as a digital scan. One or more elastic positioning dental appliances can be fabricated from digital data that is created based on the three-dimensional virtual model.

(18) Frequently a patient's physical teeth have interproximal spaces between them—either naturally occurring or intentionally created by the doctor—which a fully successful course of treatment will close, significantly reduce or even enlarge (as in the case for making space for restorative dental treatments). Typically, an interproximal space between two adjacent physical teeth that is smaller than approximately 0.25 millimeters (mm) may be too small to be represented in a digital scan obtained, for example, as set out in paragraph (18).

(19) Therefore, the interproximal space may not be properly represented in a conventional three dimensional (3D) virtual model that is created based on the digital scan. The digital teeth in a conventional three dimensional (3D) virtual model that correspond to the adjacent physical teeth may appear to be connected, i.e., have no interproximal space. Various embodiments are provided for creating one or more digital tooth models that more accurately depict one or more physical teeth based on interproximal information that represents an interproximal space as a part of more

successfully reducing or closing an interproximal space.

(20) In situations where an interdental space is present, the space may be sufficiently small whereby the accurate reproduction of the pre-existing space in a dental model of the teeth may be difficult or impossible. These spaces may be naturally occurring or artificially introduced through orthodontic treatment, for example, or through mechanical means such as a dental procedure whereby the teeth are reshaped.

(21) For example, in situations where dental crowding is present, or when certain teeth are not the correct/desired shape, recontouring of the teeth may be desirable (also called “interproximal reduction” or “IPR” for short). This can be accomplished through any number of dental instruments, including abrasive strips, discs, and/or burs. The result is that the teeth are narrowed and the resulting space in between the teeth is known as an interproximal space. An interproximal space may also be created by orthodontically moving adjacent teeth away from each other.

(22) Depending on if the gum tissue, which normally occupies the undercut between adjacent teeth is present (also known as a papilla), the interproximal space may either be a vertical space with relatively uniform width from the occlusal-most portion of the contact to the gums, or in the case of missing papilla, a vertical space at the occlusal-most portion of the contact which opens up into a triangular space near the gum line due to the undercuts of the adjacent teeth and the missing papilla.

(23) When small spaces are created, for example width-wise, or if teeth are obstructing the space due to crowding arrangements, trying to accurately determine the dimensions of the space using scans or impressions of the teeth can be particularly challenging. This is because, in order for the dimensions to be accurately captured, the impression material needs to flow in between the teeth and not be torn upon removal of the impression. If the material which flows in between the teeth becomes torn, then the resulting model of the teeth will appear as if the teeth are actually touching, when in fact, a space is present in between.

(24) For digital scans of the teeth, small spaces may be difficult to accurately capture because the scan is unable to properly characterize the areas where a direct line of sight cannot be obtained as a result of obstruction from the tooth structure being scanned.

(25) Therefore, according to one embodiment, a way to accurately capture the width and shape of the interproximal spaces (if any) is provided so that what is present in the mouth of the patient can be more accurately reproduced in a digital reconstruction of the patient's teeth on the computer.

(26) FIG. 1 illustrates an example of a patient's physical teeth **100**, according to one embodiment. For the sake of simplicity, FIG. 1 depicts a few of the patient's physical teeth **100a**, **100b**, **100c**, **100d** instead of the entire set of the patient's physical teeth **100**. As depicted in FIG. 1, there is a space **110** between the physical teeth **100b**, **100c** that are adjacent. According to one embodiment, at least a portion of the space **110** is a minimal space where the adjacent physical teeth **100b**, **100c** are closest to each other. The space **110** is also referred to as an interproximal space **110** and the physical teeth **100b**, **100c** that are adjacent are also referred to as adjacent physical teeth **100b**, **100c**. The interproximal space **110** may be naturally occurring or may be artificially created. The physical teeth **100b**, **100c** have contours **130a**, **130b** in the area **120**. The area between the contours **130a**, **130b** of adjacent teeth **100b**, **100c** defines the area **120** when the teeth are aligned along an arch.

(27) When teeth are properly aligned along an arch as depicted in FIG. 1, the interproximal space is defined by the narrowest distance between a mesial and distal surface of adjacent teeth **100b** and **100c**. However, if an adjacent tooth is rotated, as is commonly the case when teeth are not properly aligned, the interproximal space may be defined by the narrowest space between any surface of two adjacent teeth (e.g., mesial, distal, buccal, lingual or more commonly a contour portion).

(28) Sometimes dental personnel artificially create an interproximal space **110** between aligned adjacent physical teeth **100b**, **100c** in order to facilitate treatment. One way to artificially create an interproximal space **110** is interproximal reduction. As depicted in FIG. 1, the interproximal space

110 is, at least in part, artificially created due to interproximal reduction, and, therefore, the physical tooth **100c**'s contour **130b** has a flattened surface. An interproximal space **110** may be artificially created using other methods such as expansion, retraction, and moving one or more teeth, for example, to avoid extracting teeth when the patient's jaw is too small for their teeth to fit in.

(29) FIG. 2 illustrates an example of a digital teeth model **200** that represents the patient's set of physical teeth **100** (FIG. 1), according to one embodiment. The digital teeth model **200** can be obtained with a digital scan, as described in paragraph (18) hereof. Many types of imaging or scanning may be used to obtain a digital teeth model **200**. One example of an intraoral scanner that may be used is the Itero brand scanner by Cadent.

(30) The digital teeth model **200** includes digital teeth **200a-200d** that represents each of the patient's physical teeth **100** (FIG. 1). Each of the digital teeth **200a-200d**, according to one embodiment, represents one physical tooth **100a-100d**. For example, digital tooth **200a** represents physical tooth **100a**, digital tooth **200b** represents physical tooth **100b**, digital tooth **200c** represents physical tooth **100c**, and digital tooth **200d** represents physical tooth **100d**. The adjacent digital teeth **200b**, **200c** respectively represent the adjacent physical teeth **100b**, **100c**.

(31) The current technology for obtaining a digital teeth model **200** using a digital scan, as discussed herein may not accurately depict relatively small interproximal spaces. Therefore, the space **110** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1) may not be properly/accurately depicted in the digital teeth model **200**. Instead, the adjacent digital teeth **200b**, **200c** (FIG. 2) appear to be at least partially connected, for example, at a location **210** in the digital teeth model **200**. As will become more evident, the holes **220**, which result from missing data, may be filled in, for example, during a cleanup process causing the adjacent digital teeth **200b** and **200c** to appear connected or a rough estimated interproximal gap may be created.

(32) FIG. 3 illustrates an example of a conventional 3D virtual model **300** that was created based on a digital teeth model **200**, according to one embodiment. Each of the digital teeth **300a-300d** in the conventional 3D virtual model **300**, according to one embodiment, represents one physical tooth **100a-100d** (FIG. 1). For example, digital tooth **300a** represents physical tooth **100a**, digital tooth **300b** represents physical tooth **100b**, digital tooth **300c** represents physical tooth **100c**, and digital tooth **300d** represents physical tooth **100d**. The conventional 3D virtual model **300** includes adjacent digital teeth **300b**, **300c** that correspond to the adjacent physical teeth **100b**, **100c** (FIG. 1).

(33) The conventional 3D virtual model **300** may be created by a meshing process. Since the space **110** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1) is too small for a digital scan to recognize, the adjacent digital teeth **300b**, **300c** in the digital tooth **300a-300d** model appear to be connected due to the current meshing process filling the holes **220** (FIG. 2) resulting in a filled in interproximal space **320** between the adjacent digital teeth **300b**, **300c** of the conventional 3D virtual model **300**.

(34) FIG. 4 illustrates the use of an instrument **400** to measure an interproximal space between adjacent physical teeth, according to one embodiment. According to one embodiment, the instrument **400** is a piece of metal, plastic or other material of a known thickness. The instrument **400** can be inserted between the adjacent physical teeth **100b**, **100c** (FIG. 1). Several instruments **400** of different known thicknesses can be used. For example, instruments **400** in increments of approximately 0.1 mm may be used. When the appropriate amount of resistance results from the insertion of a particular instrument **400**, the thickness of that instrument **400** can be used as the interproximal information that represents the interproximal space **110** (FIG. 1). Each of the instruments **400** may have a label or other indicia specifying the thickness of each of the instruments **400**.

(35) FIGS. 9a-9f depict scannable objects that can be inserted between adjacent physical teeth **100b**, **100c** and used as a part of determining interproximal information that represents a space **110** between the adjacent physical teeth **100b**, **100c**, according to various embodiments. A digital teeth

model can be created by performing a digital scan on the patient's physical teeth **100** with a scannable object inserted snugly into a minimal space at the closest distance between the adjacent physical teeth **100b**, **100c**. If physical impressions are used, the object may be captured within the impression either as a “negative” of the object or as part of a “pick up” impression, whereby the object remains physically embedded in the impression (in the same position where originally placed on the teeth) when the impression is removed. In this regard, the object is scannable indirectly whether as a negative geometry or as picked up via the impression. The scannable object can be digitally removed from the digital teeth model to obtain all, or a part of the interproximal information based on information related to the scannable object.

(36) FIG. **9a** depicts scannable objects **900a-1**, **900a-2** with a portion **910a-1**, **910a-2**, that is shim shaped with different thicknesses along the shim shaped portion **910a-1**, **910a-2**. The scannable objects **900a-1**, **900a-2** also have non-shim shaped portions **920a-1**, **920a-2**.

(37) FIG. **9b** depicts a scannable object **900b** with a portion **910b** that is shim shaped with different thicknesses along the shim shaped portion **910b**. Although FIGS. **9a** and **9b** depict sides of the shim shaped portions **910a-1**, **910b-2**, **910b** that are symmetrical with respect to each other, various embodiments are well suited for the respective sides of the shim shaped portions **910a-1**, **910b-2**, **910b** to not be symmetrical.

(38) FIG. **9c** depicts a scannable object **900c** that is a triodent V-wedge, according to one embodiment. The scannable object **900c** has a V shaped portion **920c** at the cross section indicated by the line, a curved portion **930c**, and a handle **910c**. The V shaped portion **920c**, according to one embodiment, is a shim shaped portion with different thicknesses along the shim shaped portion. The curved portion **930c** can be used as a part of approximating a curvature of contours associated with the adjacent physical teeth. The handle **910c** can have information describing parameters associated with the scannable object **900c**, such as the height, width, lateral curvature, base curvature, the point of the curvature and the radius of the curvature. The parameters can be used as a part of determining dimensional information associated with a space **110**.

(39) FIG. **9d** depicts a scannable object **900d** that is a Triodent “fender” inserted between adjacent physical teeth, according to one embodiment. The scannable object **900d** includes a metal shim shaped portion **910d** located at the apex **920d** of a V shaped portion **930d**. According to one embodiment, the V shaped portion **930d** is a shim shaped portion with different thicknesses along the shim shaped portion. According to one embodiment, the scannable object **900d** includes a curved portion **940d**. The curved portion **940d** can be used as a part of determining a curvature of contours associated with the adjacent physical teeth as depicted in FIG. **9d**.

(40) FIG. **9e** depicts a scannable object **900e** that has notches **910e**, according to one embodiment. FIG. **9f** depicts a scannable object **900f** that has a shim shaped portion **910f** and a non shim shaped portion **920f**, according to one embodiment. The scannable objects **900e** **900f** depicted respectively in FIGS. **9e** and **9f** include respective indicia, such as notches **910e** or lines **930f**, that can be used for determining one or more dimensions associated with a space **110**. According to one embodiment, the scannable object **900c** is a rigid object. According to one embodiment, any one or more of the scannable objects **900a-900g** can be translucent. For example, assuming that the scanning device's line of sight is located on one side of the translucent scannable object **900a-900g**, the digital scan can at least approximate a tooth located on the other side of the translucent scannable object **900a-900g**. However, various embodiments are well suited for a scannable objects **900a-900g** that is opaque or semi-translucent, among other things.

(41) According to one embodiment, a scannable object as depicted in FIGS. **9a-9f** can have indicia that can be used for determining one or more dimensions associated with a space **110**. For example, information pertaining to the geometry of a scannable object can be associated with a handle **910c** (FIG. **9c**). In another example, notches **910e** (FIG. **9e**) or lines **930f** (FIG. **9f**) can be used to determine one or more dimensions of a space **110**. Examples of a dimension include but are not limited to a width or a height associated with the space **110**. A dimension may be a width associated

with the minimal space based on the thickness of a portion of the scannable object. A dimension may be a height of an interproximal area that results from interproximal reduction of at least one of the adjacent physical teeth. The indicia can be used to determine one or more dimensions from a digital teeth model of the set of physical teeth taken with the scannable object inserted into the space **110**, as will become more evident.

(42) FIGS. **10a-10c** depict scannable objects inserted between adjacent physical teeth, according to various embodiments.

(43) FIG. **10a** depicts several scannable objects **900b** inserted between the adjacent physical teeth to more accurately determine the shape or size of the respective spaces **1010a**, **1020a**, **1030a**. The spaces **1010a**, **1020a**, **1030a** between various adjacent physical teeth in this illustration are triangular shaped. Each of the scannable objects **900b** depicted in FIG. **10a** may be different sizes selected to best fit the respective space **1010a**, **1020a**, **1030a** that they are inserted into.

(44) FIG. **10b** depicts a scannable object **1010b** for determining a height associated with a space between adjacent physical teeth. The scannable object **1010b** has indicia **1020b** as described herein. The indicia **1020b** can be used to determine a height **1040b** of an interproximal area **1030b** that resulted from interproximal reduction of at least one **1050b** of the adjacent physical teeth. The indicia **1020b** can be used to determine a height **1060b** between the top **1070b** of at least one of the adjacent physical teeth and the papilla **1080b** of the gingival between the adjacent physical teeth.

(45) FIG. **10c** depicts a scannable object **900f** (FIG. **9f**) inserted between adjacent physical teeth that can also be used for determining a dimension associated with the space between the adjacent physical teeth. For example, the scannable object **900f** can be used for determining a width associated with the minimal space based on the thickness of a portion of the scannable object **900f**, determining a height of an interproximal area that results from interproximal reduction of at least one of the adjacent physical teeth, or determining a height between the top of at least one of the adjacent physical teeth and the papilla of the gingiva between the adjacent physical teeth, as discussed herein.

(46) A first digital teeth model can be created by digitally scanning the set of physical teeth with a scannable object (FIGS. **9a-9f**) snugly inserted into the minimal space between the adjacent physical teeth and the scannable object can be digitally removed from the first digital teeth model to determine all or part of the interproximal based on information related to the scannable object. The scannable object can be removed, according to one embodiment, by receiving a second digital teeth model of the set of physical teeth of the patient without the scannable object inserted into the space and moving segmented teeth of the second digital teeth model to coincide with locations of the teeth of the first digital model.

(47) A scannable object (FIGS. **9a-9f**) can be used for determining the closest distance between the adjacent physical teeth, a spatial orientation of the adjacent physical teeth relative to the closest distance and the location of the closest distance. For example, referring to FIG. **10a**, the positions of the scannable objects **900b** reflect the spatial orientation of the respective physical teeth that the scannable objects **900b** are between. The scannable objects **900b** can be inserted snugly into the respective spaces **1010a**, **1020a**, **1030a** to determine the locations of the closest distances between the respective adjacent physical teeth and to determine the spatial orientations relative to the closest distances.

(48) One or more dimensions can be determined based on indicia associated with the scannable object (FIGS. **9a-9f**) in the first digital teeth model, as discussed herein. Indicia associated with an exposed portion of the inserted scannable object can be used to determine the one or more dimensions. Various embodiments are well suited to dimensions besides just width or height information. Various embodiments are well suited for any dimensional information that enables more accurately model the space between the adjacent physical teeth. For example, the dimensional information may include shapes or sizes, or a combination thereof, among other things, of various portions or areas of the space. The dimensional information may be three-dimensional (3D)

information. Scannable objects with curved shaped portions can be used to more accurately model curved contours and scannable objects with shim shaped portions can be used to more accurately model triangularly shaped contours, among other things as discussed herein. The best fitting scannable object, according to one embodiment, is selected for insertion to more accurately capture one or more dimensions associated with the space.

(49) FIG. 5 illustrates an example of a user interface **500**, according to one embodiment. FIG. 5 is an example of a user interface **500** that is described by way of illustration and not by way of limitation. According to one embodiment, the user interface **500** is a part of a system that performs digital scans.

(50) As illustrated in FIG. 5, the user interface **500** depicts representations of teeth. Dental personnel, such as a doctor or a technician, can specify interproximal information using the user interface **500**. For example, the dental personnel can specify a measurement between adjacent physical teeth, identification of the adjacent physical teeth, and identification of any physical teeth with flattened surfaces due to interproximal reduction, among other things. The adjacent physical teeth can be specified with respect to the other physical teeth. As depicted in FIG. 5, adjacent physical teeth with respective interproximal spaces between them have been specified by circling **510** between the adjacent physical teeth. Other methods of specifying adjacent physical teeth can be used. For example, the name or location, or a combination thereof can be used to specify adjacent physical teeth. The user interface **500** may receive information that an interproximal reduction is planned to be or has been performed on at least one of the adjacent physical teeth.

(51) Although the user interface **500** illustrated in FIG. 5 depicts representations of teeth, various embodiments are well suited to a user interface **500** for specifying interproximal information without depicting representations of teeth. For example, other methods of specifying adjacent physical teeth can be used. For example, the user interface may provide fields or drop-down menus, among other things, for specifying the names of adjacent physical teeth or locations of adjacent physical teeth, the amount and/or special relationship of the interproximal space or a combination thereof.

(52) Alternatively, or in addition to using the user interface **500**, the dental personnel may submit a digital or physical form specifying the appropriate information, or a subset thereof, when sending the digital scan, for example, to a facility that creates digital tooth models, according to various embodiments. Other methods besides a user interface **500**, such as a form, may also be used for specifying that an interproximal reduction is planned to be or has been performed on at least one of the adjacent physical teeth.

(53) The specified interproximal information can be used as a part of a segmentation process that creates digital tooth models **600** (FIG. 6) from the digital teeth model **200** (FIG. 2) while maintaining the interproximal space, according to one embodiment.

(54) FIG. 6 illustrates an example of created digital tooth models **600**, according to one embodiment. According to one embodiment, the created digital tooth models **600** are three dimensional models of the patient's physical teeth **100** (FIG. 1). Each of the digital tooth models **600**, according to one embodiment, represents one physical tooth **100** (FIG. 1). For example, digital tooth model **600a** represents physical tooth **100a**, digital tooth model **600b** represents physical tooth **100b**, digital tooth model **600c** represents physical tooth **100c**, and digital tooth model **600d** represents physical tooth **100d**.

(55) The digital tooth models **600** can be created based on the interproximal information that represents the space **110** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1) that correspond to the digital tooth models **600b**, **600c** (FIG. 6). According to one embodiment, the digital tooth models **600** are created as a part of segmenting the digital teeth model **200** (FIG. 2) into the digital tooth models **600** while maintaining the interproximal space **610** (FIG. 6) based on the interproximal information pertaining to a physical interproximal space **110** (FIG. 1).

(56) As discussed herein, with the conventional segmentation process, since the space **110** (FIG. 1)

between the adjacent physical teeth **100b**, **100c** (FIG. 1) is too small for a digital scan to recognize, the adjacent digital teeth **300b**, **300c** (FIG. 3) in the conventional three dimensional (3D) virtual model **300** (FIG. 3) appear to be, at least partially, connected due to the conventional meshing or other cleanup process filling the holes **2210** in the area at **210** between the adjacent digital teeth **220b**, **200c** (FIG. 2). In contrast, according to various embodiments, the segmentation process will receive interproximal information and will prevent filling the holes **210** (FIG. 2) and will use this information to help properly create the interproximal surfaces of adjacent teeth.

(57) The digital tooth models **600** more accurately depict the adjacent physical teeth **100b**, **100c** (FIG. 1) than the digital teeth **200b**, **200c** (FIG. 2) in the digital teeth model **200** (FIG. 2) due to being created based on the interproximal information, according to one embodiment. For example, since the interproximal space **110** between the adjacent physical teeth **100b**, **100c** is too small for the digital scan to recognize, the interproximal space at location **210** is not properly depicted between the adjacent digital teeth **200b**, **200c** (FIG. 2) in the digital teeth model **200** (FIG. 2). When a conventional 3D virtual model **300** (FIG. 3) is created from the digital teeth model **200** (FIG. 2), the adjacent digital teeth **300b**, **300c** (FIG. 3) in the conventional 3D virtual model **300** (FIG. 3) appear to be connected in the interproximal space **320** (FIG. 3) between the digital teeth **300b**, **300c**. In contrast, according to various embodiments, interproximal information is obtained prior to or as part of performing the segmentation process on the digital teeth model **200** (FIG. 2) to maintain the interproximal space **610** (FIG. 6) between respectively created digital tooth models **600** (FIG. 6).

(58) Further, the exemplary created digital tooth models **600** can more accurately depict the contours or flattened surfaces (where IPR has been performed) of the area of the adjacent physical teeth. For the sake of illustration, assume that the flattened surface (FIG. 6) of contour **630a** (FIG. 6) corresponds to the flattened surface of contour **130a** (FIG. 1) and the contours **630a**, **630b** (FIG. 6) correspond to the contours **130a**, **130b** (FIG. 1). The digital tooth model **600c** can more accurately depict a flattened surface associated with contour **130b** of the corresponding physical tooth **100c** (FIG. 1), which was caused by interproximal reduction on that physical tooth **100c**, than a corresponding surface associated with the digital tooth **200c** (FIG. 2). Further, the contours **630a**, **630b** (FIG. 6) more accurately depict the contours **130a**, **130b** (FIG. 1) than corresponding surfaces associated with the digital teeth **200b**, **200c** (FIG. 2).

(59) According to one embodiment, the contour of the tooth in this context is “visible contour” of the tooth when viewed from specified direction. In particular for the correct reconstruction of the tooth contours, the contour is the visible contour viewed from direction perpendicular to the dental arch and parallel to the occlusal plane. Due to possible imprecision of a digital scan, the interproximal space **320** between the teeth **300b** and **300c** may be filled and some of the contours are missed in the vicinity of the interproximal space **320** (FIG. 3). According to one embodiment, the interproximal space and associated visible contours are restored using the interproximal information that represents the interproximal spaces as described herein. If the space is identified by the personal as corresponding to “interproximal reduction,” then areas of the adjacent teeth can be restored on the 3D model to produce straight line contours and flat areas. If the space is identified as “natural”, then areas of the adjacent teeth can be restored on the 3D model to produce natural contours and convex areas. In both cases though the amount of space is equal to the distance measured on physical teeth, according to one embodiment. The contours, if natural, can be obtained by many means including using information of or from: the contours of other of the patient's teeth, based on photographs of the patient's teeth, contours of standard teeth, a database of teeth contour information, or the contours can be created by a technician.

(60) According to one embodiment, an elastic positioning dental appliance (also known as “an aligner”) for realigning teeth can be created based on the created digital tooth models **600**. Such an appliance may be comprised of a thin shell of elastic material that generally conforms to a patient's physical teeth but each appliance to be used at a treatment stage has a cavity geometry that is

slightly out of alignment with the teeth arrangement at the start of that treatment stage. Placement of the elastic positioning dental appliance over the physical teeth applies controlled forces in specific locations to gradually move the physical teeth into a new arrangement.

(61) A series of aligners can be used to move the physical teeth through a series of intermediate arrangements to a final desired arrangement. Due to the limited space within the oral cavity and extensive movements that some physical teeth may typically undergo for treatment, at least some of the physical teeth will often be moved throughout the series of intermediate arrangements to properly arrange the physical teeth. Thus, a single patient treated with elastic positioning dental appliance may experience from 2 to over 50 stages (with an average of 25-30 aligner stages per arch) before achieving the final desired teeth arrangement.

(62) According to one embodiment, the digital teeth model **200** (FIG. 2) is segmented into digital tooth models **600** (FIG. 6) so that the digital tooth models **600** (FIG. 6) can be based on the movement of the physical teeth **100** (FIG. 1) through the series of intermediate patterns due to the course of treatment. Each of the digital tooth models **600** (FIG. 6) may have one or more axes of its own and three dimensional (3D) coordinates so that each of the digital tooth models' **600** position can be based on the position of a corresponding physical tooth **100** (FIG. 1) at a given point in time, as will be described in more detail in the context of FIG. 11. By moving the digital tooth models **600** (FIG. 6), intermediate aligners can be fabricated for each of the series of intermediate patterns of physical teeth **100** (FIG. 1) movement.

(63) Various embodiments are provided for creating one or more digital tooth models **600** (FIG. 6) that more accurately depict one or more physical teeth **100** (FIG. 1) based on interproximal information that represents the interproximal space **110** (FIG. 1). Thus, an aligner or a series of aligners that are created based on digital tooth models **600** (FIG. 6) that more accurately depict the physical teeth **100** (FIG. 1), according to various embodiments, can more successfully and more easily properly align the teeth and close or significantly reduce or enlarge, as desired, the interproximal spaces **110** (FIG. 1) between the patient's teeth **100** (FIG. 1).

(64) More information pertaining to the planning and fabrication of aligners as dental appliances is described in detail in U.S. Pat. No. 5,975,893, and in published PCT application WO 98/58596 which designates the United States and which is assigned to the assignee of the present application.

(65) FIG. 7 illustrates a system **700** for creating a digital tooth model **600** (FIG. 6) of a patient's tooth **100** (FIG. 1) using interproximal information, according to one embodiment. The blocks that represent features in FIG. 7 can be arranged differently than as illustrated and can implement additional or fewer features than what are described herein. Further, the features represented by the blocks in FIG. 4 can be combined in various ways. The system **700** can be implemented using hardware, hardware and software, hardware and firmware, or a combination thereof. According to one embodiment, the system **700** is part of a system that performs segmentation, as discussed herein.

(66) The system **700** can include an interproximal-information-obtaining-component **710**, a digital-teeth-model-receiving-component **720**, and a digital-tooth-model-creation-component **730**. The system **700** may also include an optional physical-tooth-specification-component **712**.

(67) The interproximal-information-obtaining-component **710** is configured for receiving interproximal information that represents a space **110** (FIG. 1) between adjacent physical teeth **100b**, **100c** (FIG. 1) of the patient. The space **110** (FIG. 1) may represent a naturally occurring space between the adjacent physical teeth **100b**, **100c** or represent an artificially created space between the adjacent physical teeth **100b**, **100c**.

(68) The digital-teeth-model-receiving-component **720** is configured for receiving a digital teeth model **200** (FIG. 2) of a set of physical teeth **100** (FIG. 1) of the patient that includes the adjacent physical teeth **100b**, **100c** (FIG. 1). The digital teeth model **200** (FIG. 2) can be obtained from a digital scan as described above of the set of physical teeth **100** (FIG. 1).

(69) The digital-tooth-model-creation-component **730** is configured for creating one or more digital

tooth models **600a-600d** (FIG. 6) that more accurately depicts one or more of the physical teeth **100** (FIG. 1) than corresponding digital teeth **200a-200d** (FIG. 2) included in the digital teeth model **200** (FIG. 2) based on the interproximal information.

(70) The digital-tooth-model-creation-component **730** may be further configured for creating digital tooth models **600b**, **600c** (FIG. 6) representing the adjacent physical teeth **100b**, **100c** (FIG. 1), which more accurately depict contours **130a**, **130b** (FIG. 1) of the adjacent physical teeth **100b**, **100c** (FIG. 1) in area **120** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1), for creating a digital tooth model **600c** (FIG. 6) that more accurately depicts a flattened surface of contour **130b** of a physical tooth **100c** (FIG. 1) due to interproximal reduction, or for creating a digital tooth model **600c** (FIG. 6) based on information indicating that an interproximal reduction is planned to be or has been performed on at least one of the adjacent physical teeth **100b**, **100c** (FIG. 1), or a combination thereof.

(71) The system **700** may also include a physical-tooth-specification-component **712** configured for receiving information specifying which of the set of physical teeth **100** (FIG. 1) the adjacent physical teeth **100b**, **100c** (FIG. 1) are. According to one embodiment, the information specifying which of the set of physical teeth **100** (FIG. 1) the adjacent physical teeth **100b**, **100c** (FIG. 1) are, is a part of the interproximal information and may be received by the interproximal-information-obtaining-component **710**. The physical-tooth-specification-component **712** may be a part of the interproximal-information-obtaining-component **710** or may communicate with the interproximal-information-obtaining-component **710**.

(72) The system **700** can include one or more computer processors for performing the operations of receiving of the interproximal information, the receiving of the digital teeth model **200** (FIG. 2) and the creating of the one or more digital tooth models **600** (FIG. 6).

(73) FIG. 8 illustrates a method of creating a digital tooth model of a patient's tooth using interproximal information, according to one embodiment.

(74) At **810**, the method begins.

(75) At **820**, interproximal information is received that represents a space **110** (FIG. 1) between adjacent physical teeth **100b**, **100c** (FIG. 1) of the patient. For example, an instrument **400** (FIG. 4) can be inserted between the adjacent physical teeth **100b**, **100c** (FIG. 1). Several instruments **400** of different known widths can be used. When the appropriate amount resistance results from the insertion of a particular instrument **400**, the width of that instrument **400** can be used as the interproximal space **110** (FIG. 1). Each of the instruments **400** may have a label or other indicia (such as numbers, shape, colors, letters, notches, etc.), including geometric indicia, specifying the thickness of each of the instruments **400**.

(76) The thickness associated with the instrument **400** that provides appropriate resistance can be entered into a user interface **500** (FIG. 5). Further, the adjacent physical teeth **100b**, **100c** (FIG. 1) can be specified by a user interface **500**. The user interface **500** may be a part of a system that performs digital scans which may be used by a treating professional, lab, service provider or manufacturer.

(77) Various embodiments are also well suited for receiving interproximal information that represents a space **110** obtained, at least in part, by inserting a scannable object (FIGS. 9a-9f) in the space **110**, as described in the context of FIGS. 9a-10c.

(78) According to one embodiment, information is received that specifies the adjacent physical teeth **100b**, **100c** (FIG. 1) with respect to other physical teeth **100a**, **100d** (FIG. 1) associated with the set of physical teeth **100** (FIG. 1). For example, the interproximal information can also include identification of the adjacent teeth **100b**, **100c** (FIG. 1) that were specified by circling **510** (FIG. 5) representations of teeth displayed on a user interface **500**, identifications of the adjacent teeth **100b**, **100c** (FIG. 1), or names of the adjacent teeth **100b**, **100c** (FIG. 1), among other things.

(79) The received interproximal information can represent a naturally occurring space **110** between the adjacent physical teeth **100b**, **100c** (FIG. 1) or an artificially created space **110** between the

adjacent physical teeth **100b**, **100c** (FIG. 1).

(80) Information can be received indicating that an interproximal reduction is planned to be or has been performed on at least one of the adjacent physical teeth **100b**, **100c** (FIG. 1). This information may be a part of or separate from the received interproximal information.

(81) The interproximal information, according to one embodiment, is obtained prior to the segmentation process, for example, by a system that performs digital scans. The interproximal information may be communicated to an interproximal-information-obtaining-component **710** of the system **700** (FIG. 7). The system **700** may be a part of a segmentation system.

(82) At **830**, a digital teeth model **200** (FIG. 2) of a set of physical teeth **100** (FIG. 1) of the patient that includes the adjacent physical teeth **100b**, **100c** (FIG. 1) is received. For example, a digital teeth model **200** (FIG. 2) of a set of physical teeth **100** (FIG. 1) that includes adjacent physical teeth **100b**, **100c** (FIG. 1) can be received by a digital-teeth-model-receiving-component **720** of the system **700** (FIG. 7).

(83) The received digital teeth model **200** (FIG. 2) can be derived from a digital scan of physical teeth **100** (FIG. 1), as described above. The interproximal information received at **820** may be included in a digital teeth model **200** (FIG. 2) that is received at **830**.

(84) At **840**, one or more digital tooth models **600** (FIG. 6) is created that more accurately depicts one or more of the physical teeth **100a-100d** (FIG. 1) than corresponding digital teeth **200a-200d** (FIG. 2) included in the digital teeth model **200** (FIG. 2) based on the interproximal information. For example, one or more digital tooth models **600a-600d** (FIG. 6) can be created by a digital-tooth-model-creation-component **730** of the system **700** (FIG. 7). The digital tooth models **600** (FIG. 6) can be created based on the interproximal information that represents the space **110** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1) that correspond to the digital tooth models **600b**, **600c** (FIG. 6).

(85) The digital tooth models **600b**, **600c** (FIG. 6) more accurately depict the adjacent physical teeth **100b**, **100c** (FIG. 1) than the digital teeth **200b**, **200c** (FIG. 2) in the digital teeth model **200** (FIG. 2) due to being created based on the interproximal information, according to one embodiment. For example, since the interproximal space **110** (FIG. 1) between the adjacent physical teeth **100b**, **100c** (FIG. 1) may be too small for a digital scan to recognize, the interproximal space **110** is not properly/accurately depicted between the adjacent digital teeth **200b**, **200c** (FIG. 2) in the digital teeth model **200** (FIG. 2). The area associated with the interproximal space **110** between the adjacent digital teeth **200b**, **200c** (FIG. 2) may be, at least partially, connected with some holes **220**. When a conventional 3D virtual model **300** (FIG. 3) is created from the digital teeth model **200**, the holes **220** (FIG. 2) in the digital teeth model **200** may be filled causing the adjacent digital teeth **300b**, **300c** (FIG. 3) in the conventional 3D virtual model **300** (FIG. 3) to be connected in the vicinity of the interproximal space **320** (FIG. 3). In contrast, according to various embodiments, interproximal information is obtained prior to or as part of performing the segmentation process on the digital teeth model **200** (FIG. 2) and can be used as a part of the segmentation process to maintain the interproximal space **110** (FIG. 1) as depicted at interproximal space **610** (FIG. 6) between respectively created digital tooth models **600b**, **600c** (FIG. 6). According to one embodiment, the interproximal space **610** is an interproximal space model.

(86) Further, the exemplary created digital tooth models **600** (FIG. 6) can more accurately depict naturally occurring or, at least partially created, contours of the area of the adjacent physical teeth. Assume for the sake of illustration that contours **630a**, **630b** (FIG. 6) correspond to contours **130a**, **130b** (FIG. 1) and that area **620** (FIG. 6) corresponds to area **120** (FIG. 1). In this case, the contours **630a**, **630b** (FIG. 6) of the digital tooth models **600b**, **600c** (FIG. 6) can more accurately depict the contours **130a**, **130b** (FIG. 1) of the interproximal area **120** (FIG. 1) than corresponding surfaces associated with corresponding digital teeth **300b**, **300c** (FIG. 3) of the conventional 3D virtual model **300** (FIG. 3).

(87) The digital tooth model **600c** (FIG. 6) can more accurately depict the flattened surface of the corresponding physical tooth that was caused by interproximal reduction on that physical tooth. Assume for the sake of illustration that the flattened surface associated with contour **630b** (FIG. 6) of digital model **600c** (FIG. 6) corresponds to the flattened surface associated with contour **130b** (FIG. 1) of the physical tooth **100c** (FIG. 1). In this case, the flattened surface of the digital tooth model **600c** (FIG. 6) can more accurately depict the flattened surface of the physical tooth **100c** (FIG. 1) than a corresponding surface of digital tooth **300c** (FIG. 3) of the conventional 3D virtual model **300** (FIG. 3).

(88) At **850**, the method ends.

(89) The receiving of the interproximal information at **820**, the receiving of the digital teeth model at **830** and the creating of the one or more digital tooth models at **840** can be performed, for example, by one or more computer processors associated with a system **700** (FIG. 7).

(90) According to one embodiment, a dental appliance or a series of dental appliances can be fabricated based on the digital tooth models **600** (FIG. 6) that more accurately depict the patient's physical teeth **100** (FIG. 1) based on the interproximal information. A dental appliance or a series of dental appliances that more accurately depict the patient's physical teeth **100**, according to various embodiments, can more successfully and more easily align teeth and may close or significantly reduce or enlarge, as desired, an interproximal space **110** (FIG. 1) between the patient's teeth **100b**, **100c** (FIG. 1).

(91) According to one embodiment, a digital teeth model **200** (FIG. 2) is segmented into digital tooth models **600a-600d** (FIG. 6) so that the digital tooth models **600a-600d** can be used as a basis for the movement of corresponding physical teeth **100a-100d** (FIG. 1) through the series of intermediate arrangements during the course of treatment. Each of the digital tooth models **600a-600d** may have one or more axes of its own and three dimensional (3D) coordinates so that each of the digital tooth models' **600a-600d** can be freely positioned in 3D space based on the position of a corresponding physical tooth **100a-100d** at a given point in time. The 3D coordinates alone or 3D coordinates in combination with one or more axes of each of the digital tooth models **600a-600d** can be used to match or closely approximate the position of each of the corresponding physical teeth **100a-100d**. Intermediate and final aligners can be fabricated based on each of the series of intermediate and final arrangements.

(92) For example, FIG. 11 depicts one tooth **1100** that may be used to match or closely approximate the position of a corresponding tooth, according to one embodiment. According to one embodiment, the digital tooth **1100** represents a segmented or partially segmented digital tooth, that has one or more axes and three dimensional (3D) coordinates so that the digital tooth **1100** can be freely positioned in 3D space. For example, the 3D coordinates x, y, and z alone or 3D coordinates x, y, and z in combination with one or more axes **1104**, **1106**, **1108** can be used for positioning the digital tooth **1100**.

(93) As a frame of reference describing how a digital tooth **1100** may be moved, an arbitrary centerline (CL) may be drawn through the digital tooth **1600**. With reference to this centerline (CL), a tooth **1100** may be moved in orthogonal directions represented by axes **1104**, **1106**, and **1108** (where **1104** is the centerline). The centerline may be rotated about the axis **1108** (root angulation) and the axis **1104** (torque) as indicated by arrows **1110** and **1112**, respectively. Additionally, the tooth **1100** may be rotated about the centerline, as represented by an arrow **1112**. Thus, all possible free-form motions of the tooth **1100** can be performed.

(94) Although specific operations are disclosed in flowchart **800**, such operations are exemplary. That is, embodiments of the present invention are well suited to performing various other operations or variations of the operations recited in flowchart **800**. It is appreciated that the operations in flowchart **800** can be performed in an order different than presented, and that not all of the operations in flowchart **800** can be performed.

(95) The above illustration is only provided by way of example and not by way of limitation. There

are other ways of performing the method described by flowchart **800**.

(96) FIGS. **12A** and **12B** depict an example of a patient's physical teeth at various stages of position as a part of performing interproximal reduction, according to various embodiments. Interproximal reduction can be used to avoid moving physical teeth along the arch to create space for other teeth to move (also referred to as “distillization”) or extraction of one or more physical teeth.

(97) The patient's set of physical teeth include physical teeth **12a-12f**. Each physical tooth **12a-12f** has a distal and a mesial surface. For example, physical tooth **12a** has a distal surface **a1** and mesial surface **a2**, physical tooth **12b** has a distal surface **b1** and mesial surface **b2**, physical tooth **12c** has a distal surface **c1** and mesial surface **c2**, physical tooth **12d** has a distal surface **d1** and mesial surface **d2**, and physical tooth **12e** has a distal surface **e1** and mesial surface **e2**.

(98) The physical teeth **12a-12f** are depicted at five different stages **90a-90e** of position. The patient's set of physical teeth **12a-12f** are depicted in relationship to the arch **80** and medial axis **70** of the patient's jaw at each stage **90a-90e**.

(99) At the initial stage **90a** of alignment, the physical teeth **12b**, **12c**, **12d** are overlapping. More specifically, the mesial surface **b2** of physical tooth **12b** is in front of the distal surface **c1** of physical tooth **12c**, the mesial surface **c2** of physical tooth **12c** is in front of the distal surface **d1** of physical tooth **12d**.

(100) At stage **90b**, physical teeth **12c**, **12d** are moved outwards to remove the overlapping of the physical teeth **12b**, **12c**, **12d**. As depicted, the physical teeth **12c**, **12d** are moved by tilting them toward the patient's lips.

(101) At stage **90c**, physical teeth **12c**, **12d** are rotated to orient the mesial surface and distal surface of adjacent physical teeth **12b**, **12c**, **12d**, **12e** in preparation of interproximal reduction at locations **60a**, **60b**, and **60c**. More specifically as depicted in this illustration, the physical teeth **12c**, **12d** are rotated to orient mesial surface **b2** toward the distal surface **c1** for adjacent physical teeth **12b**, **12c**, the mesial surface **c2** toward the distal surface **d1** for adjacent physical teeth **12c**, **12d**, and the mesial surface **d2** toward the distal surface **e1** for adjacent physical teeth **12d**, **12e**. As can be seen, the physical teeth **12b**, **12c**, **12d**, **12e** are slightly forward of the arch **80** at stage **90c** because the physical teeth **12b**, **12c**, **12d**, **12e** are still too large to be moved back into the arch **80**.

(102) According to one embodiment, interproximal reduction can be performed at locations **60a**, **60b**, and **60c** resulting in interproximal spaces **10a**, **10b**, **10c** as depicted in stage **90d**. The positions of the physical teeth **12a-12f** are the same in stages **90c** and **90d**. The interproximal spaces **10a**, **10b**, **10c** provide sufficient room to move the physical teeth **12b**, **12c**, **12d**, **12e** into the arch **80** and to properly align each of the physical teeth **12a-12f** with respect to each other as depicted in stage **90e**.

(103) Various embodiments provide for correcting scan data of physical teeth in a current position to better represent contours of physical teeth in the vicinity of an interproximal space for treatment and possibly for performing interproximal reduction. For example, referring to stage **90a** using a conventional method, the triangular areas between respective overlapping adjacent physical teeth **12b**, **12c**, **12d** may be filled with data. Further, the contours of the triangular areas on the lingual side may appear to be closer to the tongue than is the case in reality and the contours of the triangular areas on the buccal side may appear to be closer to the lips than is the case in reality. Further, the interproximal spaces between the physical teeth **12a**, **12b** and **12d** and **12e** may be filled in resulting in adjacent physical teeth **12a**, **12b** and adjacent physical teeth **12d** and **12e** appearing to be connected.

(104) In contrast, using various embodiments, various scannable objects as depicted in FIGS. **9a-9f**, can be inserted into the interproximal spaces between adjacent physical teeth and a digital scan can be taken. For example, a scannable object may be inserted between physical teeth **12c** and **12d**. Due to the overlap of the adjacent physical teeth **12c** and **12d**, the scannable object may lay in an approximately flat position on the buccal surface between physical teeth **12c** and **12d**. The scannable object can be used as a part of extrapolating contours associated with the physical teeth

12c and **12d** as discussed herein. Similarly, scannable objects can be inserted between other adjacent physical teeth depicted in FIGS. **12A** and **12B**. Although the use of various embodiments to better represent contours of physical teeth in the vicinity of an interproximal space have been discussed in the context of physical teeth **12a-12f** at stage **90a**, various embodiments are well suited for better representation of contours of physical teeth at other stages **90b-90e**.

(105) Thus, various embodiments can be used for correcting digital scan data of physical teeth in a current position to better represent contours of physical teeth in the vicinity of an interproximal space between those physical teeth.

(106) Any one or more of the embodiments described herein can be implemented using non-transitory computer readable storage medium and computer-executable instructions which reside, for example, in computer-readable storage medium of a computer system or like device.

(107) The non-transitory computer readable storage medium can be any kind of memory that instructions can be stored on. Examples of the non-transitory computer readable storage medium include but are not limited to a disk, a compact disk (CD), a digital versatile device (DVD), read only memory (ROM), flash, and so on. As described above, certain processes and operations of various embodiments of the present invention are realized, in one embodiment, as a series of instructions (e.g., software program) that reside within non-transitory computer readable storage memory of a computer system and are executed by the computer processor of the computer system. When executed, the instructions cause the computer system to implement the functionality of various embodiments of the present invention. According to one embodiment, the non-transitory computer readable storage medium is tangible.

(108) According to one embodiment, a non-transitory computer readable storage medium having computer-executable instructions stored thereon for causing a computer system to perform a method of creating a digital tooth model of a patient's tooth using interproximal information is provided. For example, interproximal information is received that represents a space **110** (FIG. **1**) between adjacent physical teeth **100b**, **100c** (FIG. **1**) associated with a set of physical teeth **100** (FIG. **1**). The space **110** can be a naturally occurring space between the adjacent physical teeth or an artificially created space. Interproximal information can be determined by manually inserting pieces of material, which each have a different thickness, between the adjacent physical teeth **100b**, **100c** (FIG. **4**). User specified information specifying the adjacent physical teeth **100b**, **100c** (FIG. **1**) with respect to other physical teeth **100a**, **100d** (FIG. **1**) associated with the set of physical teeth **100** (FIG. **1**) can be received. The received user specified information can be specified by a circle **510** (FIG. **5**) around visual representations of the adjacent physical teeth **100b**, **100c** (FIG. **1**), identification information of the adjacent physical teeth **100b**, **100c** (FIG. **1**), or locations of the adjacent physical teeth **100b**, **100c** (FIG. **1**), or a combination thereof.

(109) Further, according to one embodiment, the non-transitory computer readable storage medium provides for the creation of digital tooth models **600b**, **600c** (FIG. **6**) that more accurately depict the adjacent physical teeth **100b**, **100c** (FIG. **1**) than corresponding digital teeth **200b**, **200c** (FIG. **2**) included in a digital teeth model **200** (FIG. **2**) based on the interproximal information, according to one embodiment. For example, one or more digital tooth models **600** (FIG. **6**) can be created that include adjacent digital tooth models **600b**, **600c** (FIG. **6**) that more accurately depict contours **130a**, **130b** (FIG. **1**) of the adjacent physical teeth **100b**, **100c** (FIG. **1**). An area associated with the interproximal space **110** (FIG. **2**) of the digital teeth model **200** (FIG. **2**) that corresponds to the space **110** (FIG. **1**) between the adjacent physical teeth **100b**, **100c** (FIG. **1**) can be identified and a digital tooth model **600b**, **600c** (FIG. **6**) can be created that represents one of the adjacent physical teeth **100b**, **100c** (FIG. **1**) by preventing the digital tooth model **600b**, **600c** (FIG. **6**) from extending into an area **610** (FIG. **6**).

(110) Creating digital tooth models **600** (FIG. **6**) of a patient's teeth **100** (FIG. **1**) using interproximal information which enables more accurate digital tooth models at least provides, for example, improved fitting dental appliance, improved treatment outcome, improved closure or

opening of interproximal spaces, more confidence for the dental personnel, and the ability to better track interproximal space closure from a research and development stand point.

(111) Example embodiments of the subject matter are thus described. Although the subject matter has been described in a language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

(112) Various embodiments have been described in various combinations and illustrations.

However, any two or more embodiments or features may be combined. Further, any embodiment or feature may be used separately from any other embodiment or feature. Phrases, such as “an embodiment,” “one embodiment,” among others, used herein, are not necessarily referring to the same embodiment. Features, structures, or characteristics of any embodiment may be combined in any suitable manner with one or more other features, structures, or characteristics.

Claims

1. A method of generating a digital dental model of a patient's teeth, the method comprising: receiving a digital scan of the patient's teeth; generating a three-dimensional digital teeth model from the digital scan of the patient's teeth; calculating interproximal information of the patient's teeth based at least in part on the digital scan; mapping the interproximal information onto the three-dimensional digital teeth model; modifying the three-dimensional digital teeth model by generating, based on the mapped interproximal information, contours of one or more teeth of the three-dimensional digital teeth model; modifying one or more tooth restoration treatment stages of a restorative dental treatment using the contours of the one or more teeth of the modified three-dimensional digital teeth model; and generating instructions for fabrication of a dental appliance for transmission to a fabrication machine, wherein the fabrication machine receives the instructions and fabricates the dental appliance based at least in part on the instructions.
2. The method of claim 1, wherein the interproximal information represents one or more interproximal spaces.
3. The method of claim 2, wherein the one or more interproximal spaces correspond to interproximal reductions or curved contours.
4. The method of claim 3, further comprising restoring one or more areas of the teeth adjacent to the one or more interproximal spaces on the three-dimensional digital teeth model.
5. The method of claim 4, wherein the amount of space restored on the three-dimensional digital teeth model corresponds to the distance between a pair of adjacent teeth measured on the patient's teeth.
6. The method of claim 4, further comprising restoring the one or more areas of the teeth to generate flat areas when the one or more interproximal spaces correspond to interproximal reductions.
7. The method of claim 4, further comprising restoring the one or more areas of the teeth to generate curved areas when the one or more interproximal spaces correspond to curved contours.
8. The method of claim 7, wherein dimension, angle, or measurement of the curved contours are obtained using information other than the digital scan of the patient's teeth.
9. The method of claim 8, wherein the information other than the digital scan of the patient's teeth comprises information from the contours of other patient's teeth, photographs of the patient's teeth, contours of standard teeth, information from a teeth contour database, contours created manually, or any combination thereof.
10. The method of claim 1, further comprising generating a geometry for at least one dental appliance based at least in part on the modified three-dimensional digital teeth model.
11. A computer-readable medium comprising computer-executable instructions for generating a

digital dental model of a patient's teeth that, when executed by a processor, cause the processor to: receive a digital scan of the patient's teeth; generate a three-dimensional digital teeth model from the digital scan of the patient's teeth; calculate interproximal information of the patient's teeth based at least in part on the digital scan; map the interproximal information onto the three-dimensional digital teeth model; modify the three-dimensional digital teeth model by generating, based on the mapped interproximal information, contours of one or more teeth of the three-dimensional digital teeth model; modify one or more tooth restoration treatment stages of a restorative dental treatment using the contours of the one or more teeth of the modified three-dimensional digital teeth model; and generate instructions for fabrication of a dental appliance for transmission to a fabrication machine, wherein the fabrication machine receives the instructions and fabricates the dental appliance based at least in part on the instructions.

12. The system of claim 11, wherein the interproximal information represents one or more interproximal spaces.

13. The system of claim 12, wherein the one or more interproximal spaces correspond to interproximal reductions or curved contours.

14. The system of claim 13, wherein the instructions further cause the processor to restore one or more areas of the teeth adjacent to the one or more interproximal spaces on the three-dimensional digital teeth model.

15. The system of claim 14, wherein the amount of space restored on the three-dimensional digital teeth model corresponds to the distance between a pair of adjacent teeth measured on the patient's teeth.

16. The system of claim 14, wherein the instructions further cause the processor to restore the one or more areas of the teeth to generate flat areas when the one or more interproximal spaces correspond to interproximal reductions.

17. The system of claim 14, wherein the instructions further cause the processor to restore the one or more areas of the teeth to generate curved areas when the one or more interproximal spaces correspond to curved contours.

18. The system of claim 17, wherein dimension, angle, or measurement of the curved contours are obtained using information other than the digital scan of the patient's teeth.

19. The system of claim 18, wherein the information other than the digital scan of the patient's teeth comprises information from the contours of other patient's teeth, photographs of the patient's teeth, contours of standard teeth, information from a teeth contour database, contours created manually, or any combination thereof.

20. The system of claim 11, wherein the instructions further cause the processor to generate a geometry for at least one dental appliance based at least in part on the modified three-dimensional digital teeth model.
